



TEAM || Binary Brigade

i OBJECT-ORIENTED PROGRAMMING

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RESTAURANT ORDER MANAGMENT SYSTEM

Order what you desire !

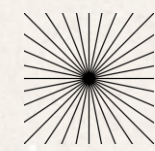
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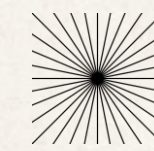
PRESENTED TO:

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INTRODUCTION



Every ones in a while we all wants to have something other than our home cooking

This project is a console-based restaurant ordering system
It allows customers and admin to use the system separately
Designed using Object Oriented Programming concepts

The system manages:

Menu

Orders

Cart

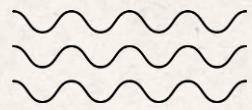
Customers

Admin controls

Objectives of the Project



- Apply OOP concepts in a real-life system
- Create a multi-user ordering system :
 - User can create multiple accounts
 - User can log in to existing accounts
 - User can log out of the system
- Allow customers to :
 - Order food
 - Checkout
- Allow admin to :
 - Manage menu
 - View customer orders



These concepts are used throughout the code.

OOP Concepts Used

Classes & Object

class MenuP

class Cart

class MenuItem

class UserSystem

class OrderItem

class AdminP

Inheritance

- Inherited base class MenuP to derived class MenuItem.

Abstraction

- abstract class MenuP
- abstract method print()

Function Overloading

- add()

Operator Overloading

Static field & method

- public static double vat
- public static double Vatcal()

Constructor features

- Constructor overloading
- Copy constructor (Deep copy)

System Features

- **Menu System Management**
 - **Cart Management**
- **Check Out & Order Storage**
- **User System (Log in, Sign up, Log out)**
 - **Admin Panel Features**

This features were used in the project system to make user experience smoother.

Menu System Management

Purpose -

- **View menu items :**
 - Customer can view the default menu items.
- **Quantity selection :**
 - Customer can choose the desired quantities of the selected item.
- **Add menu item :**
 - Customer can select and add the desired item to cart

Cart Management

Purpose -

- **Stores selected menu items :**
 - Automatically stores selected item and quantity to cart.
- **Calculates total bill**
- **Removes items :**
 - User can remove added cart item if not needed.
- **Clears cart after checkout :**
 - After customer checks out it automatically refresh the cart.

Check Out & Order Storage

Purpose-

- **Check Out Process**
 - Check out is only possible if customer is logged in.
- **Vat Calculation**
 - 5% vat is automatically added with the total bill.
- **Order Storage Logic**
 - Each customer have a separate cart storage.
- **Cart Reset After Checkout**
 - After order is stored the cart will be refreshed for next order.

User System (Log in, Sign up, Log out)

Purpose -

- **Log In :**
 - Customer can log in to system if they have existing account.
- **Sign Up :**
 - Customer can create a new account if new to system.
- **Log Out :**
 - Customer can log out of the system if they are logged in or finished using the system.

Admin Panel Features

Purpose -

- **Customer Monitoring :**
 - Admin can monitor and view customer information.
- **Order Monitoring :**
 - Admin can view the existing orders of the customers.
- **Menu Management :**
 - Admin can add and remove item from menu.
- **Encapsulation :**
 - Admin panel is strictly protected from any unauthorized excess.

Conclusion

- Built a **console-based restaurant management system** using C#
- Applied key **Object-Oriented Programming concepts**
- Implemented **menu handling, cart system, checkout, and admin panel**
- Demonstrated **real-world use of OOP** in a simple and structured way
- Strengthened understanding of **program design and code organization**

Future Improvements

- Database integration
- Graphical User Interface (GUI)
- Input validation improvements
- Order history display

*Thank
you!*

for being with us